Investigating Grover's Algorithm

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Introduction:

- Searching through unsorted data on a classical computer is slow
  - Complexity is $O(N)$, meaning every item must be checked once
- Quantum computers can search through data faster
  - **Grover's Algorithm** is considered the most efficient solution
  - Complexity is $O(\sqrt{N})$, which gives a significant speed-up

Objective:

- **Describe** the properties of several quantum gates
- **Pseudocode** Grover's algorithm
- **Simulate** Grover's Algorithm on a classical computer
- **Explore** practical applications of Grover's Algorithm

Background:

- **Qubits** store distributions of values **between 0 and 1**
  - This gives an advantage over bits, which store only **0's & 1's**
- This property allows for logical gates that are more complex than traditional classical gates, such as **AND** and **OR**
  - **Pauli-X Gate** or the quantum NOT, **swaps** the amplitude of the **$|0\rangle$ and $|1\rangle$** components
    - Represented by the (insert here) matrix
  - **Hadamard Gate** creates a **superposition** from a $|0\rangle$ or $|1\rangle$
    - This is represented by **$(1/\sqrt{2})(|0\rangle + |1\rangle)$**
    - It is also **reversible**; applying an H Gate twice returns the initial input
- Grover's Algorithm achieves its $O(\sqrt{N})$ complexity through the use of **Hadamard Gates** and **modified Z Gates**.
  - This complexity is "**asymptotically optimal**"
    - So Grover's Algorithm is the best solution! (Fig. 1)

Conclusion:

- Classical searching algorithm can be implemented in quantum
- Encryption levels need to be increased within the next ten years
  - Especially for searching algorithms that have faster speedup (Shor's Algorithm)
- Grover's Algorithm for Sudoku.
  - Grover's algorithm searches all possible Sudoku answers efficiently.
  - Grover's algorithm speeds up search by reducing the number of checks needed to find the correct answer from N to about √N.

Process & Results:

Fig. 1:

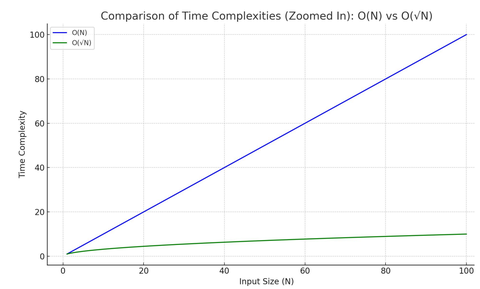


Fig. 2:

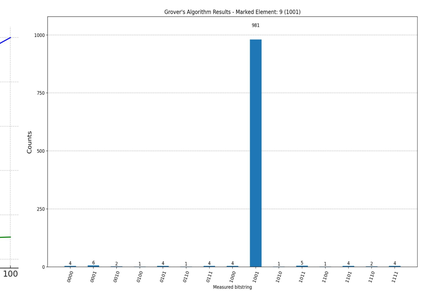
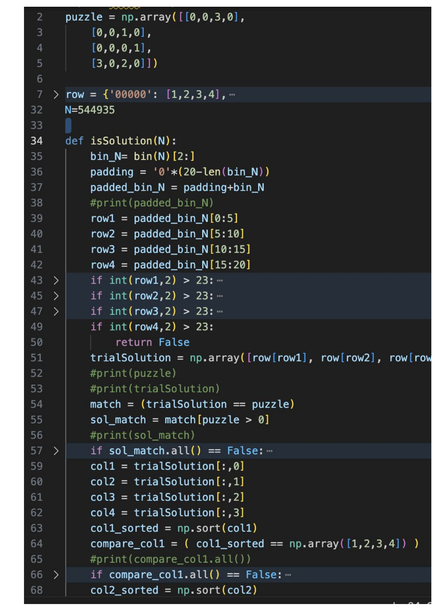


Fig. 5



```
puzzle = np.array([[0,0,3,0],
    [0,0,1,0],
    [0,0,0,1],
    [3,0,2,0]])

row = {'00000': [1,2,3,4],…
N=544935

def isSolution(N):
    bin_N= bin(N)[2:]
    padding = '0'*(20-len(bin_N))
    padded_bin_N = padding+bin_N
    #print(padded_bin_N)
    row1 = padded_bin_N[0:5]
    row2 = padded_bin_N[5:10]
    row3 = padded_bin_N[10:15]
    row4 = padded_bin_N[15:20]
    if int(row1,2) > 23:…
    if int(row2,2) > 23:…
    if int(row3,2) > 23:…
    if int(row4,2) > 23:
        return False
    trialSolution = np.array([row[row1], row[row2], row[row
    #print(puzzle)
    #print(trialSolution)
    match = (trialSolution == puzzle)
    sol_match = match[puzzle > 0]
    #print(sol_match)
    if sol_match.all() == False:…
    col1 = trialSolution[:,0]
    col2 = trialSolution[:,1]
    col3 = trialSolution[:,2]
    col4 = trialSolution[:,3]
    col1_sorted = np.sort(col1)
    compare_col1 = ( col1_sorted == np.array([1,2,3,4]) )
    #print(compare_col1.all())
    if compare_col1.all() == False:…
    col2_sorted = np.sort(col2)
```

Fig. 3:

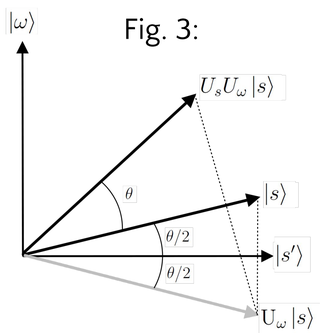


Fig. 4:

		3	
		1	
			1
3		2	

- Apply an **H Gate** to **n** qubits, giving an array with **2^n values**, all with an equal probability of being chosen

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle.$$

- Iterate the following steps $\frac{\pi}{4}\sqrt{N}$ **times**
  - Apply an **Oracle** function that multiplies **only the solution state by -1**

$$\begin{cases} U_\omega|x\rangle = -|x\rangle & \text{for } x = \omega, \text{ that is, } f(x) = 1, \\ U_\omega|x\rangle = |x\rangle & \text{for } x \neq \omega, \text{ that is, } f(x) = 0. \end{cases}$$

  - **Amplify** the magnitude of the flipped state
    - This manifests as a **reflection** around the **previous vector**
    - Apply H and X gates to all qubits, apply a multi-controlled Z gate on target qubit, then undo the H and X gates
- After all iterations for large datasets, Grover's algorithm has a chance to succeed $\mathbf{P_{success} > 1 - (1/N)}$
  - This chance becomes **extremely large** as N increases (Fig. 2)

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